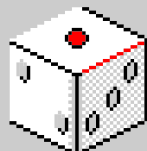
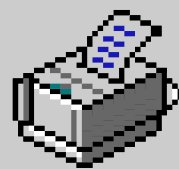
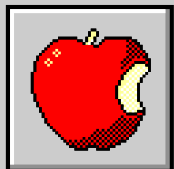
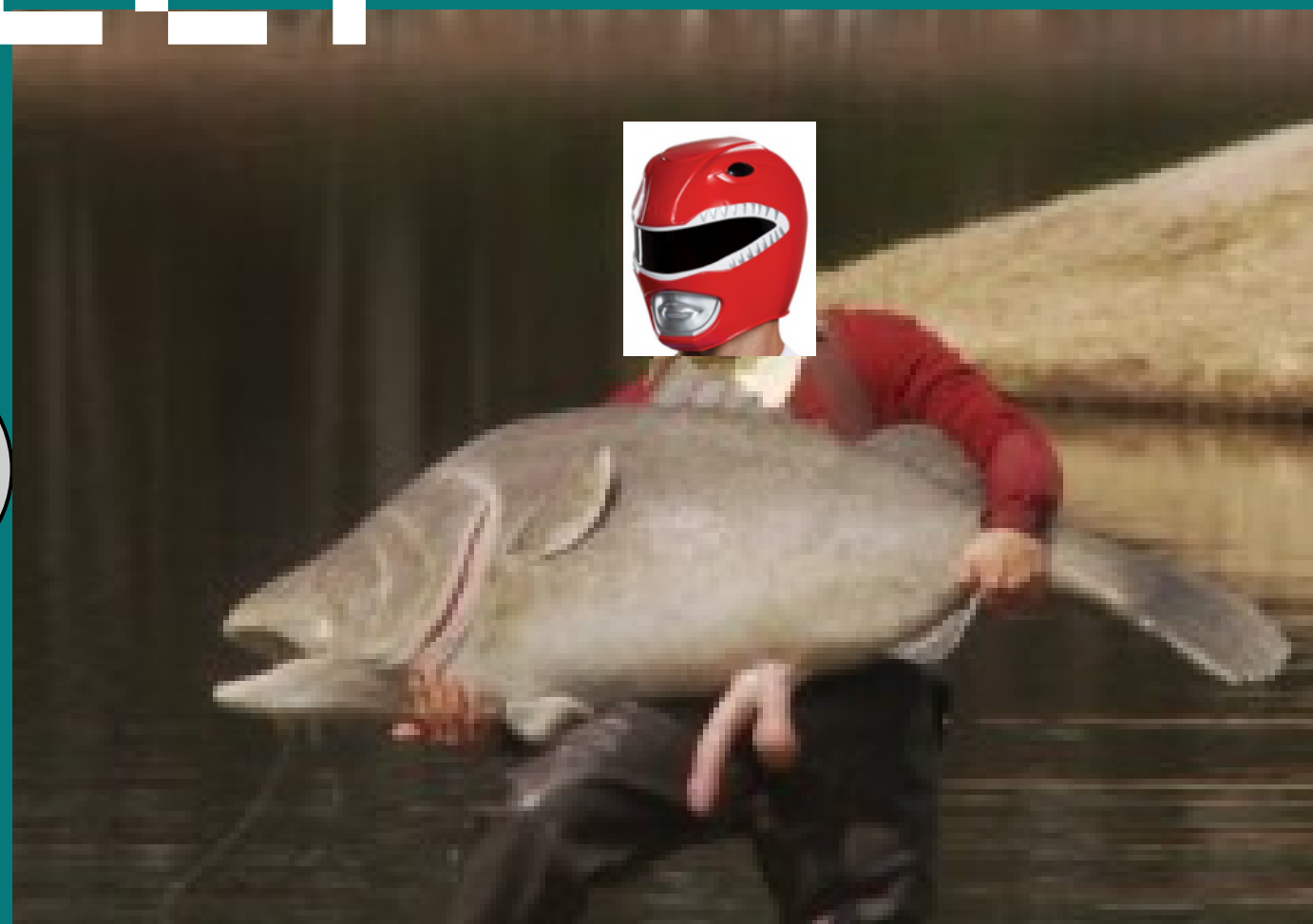


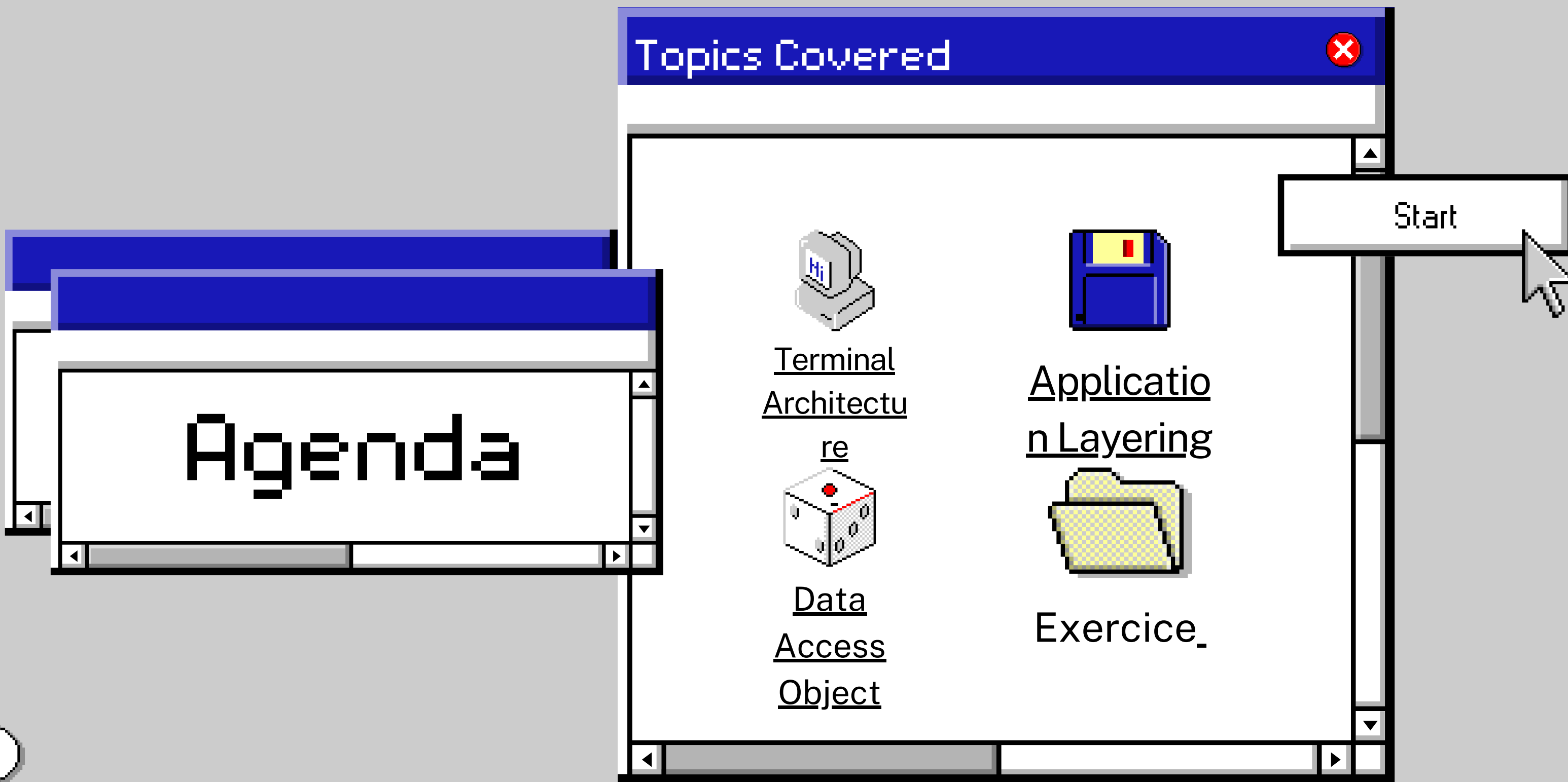
Summarizer



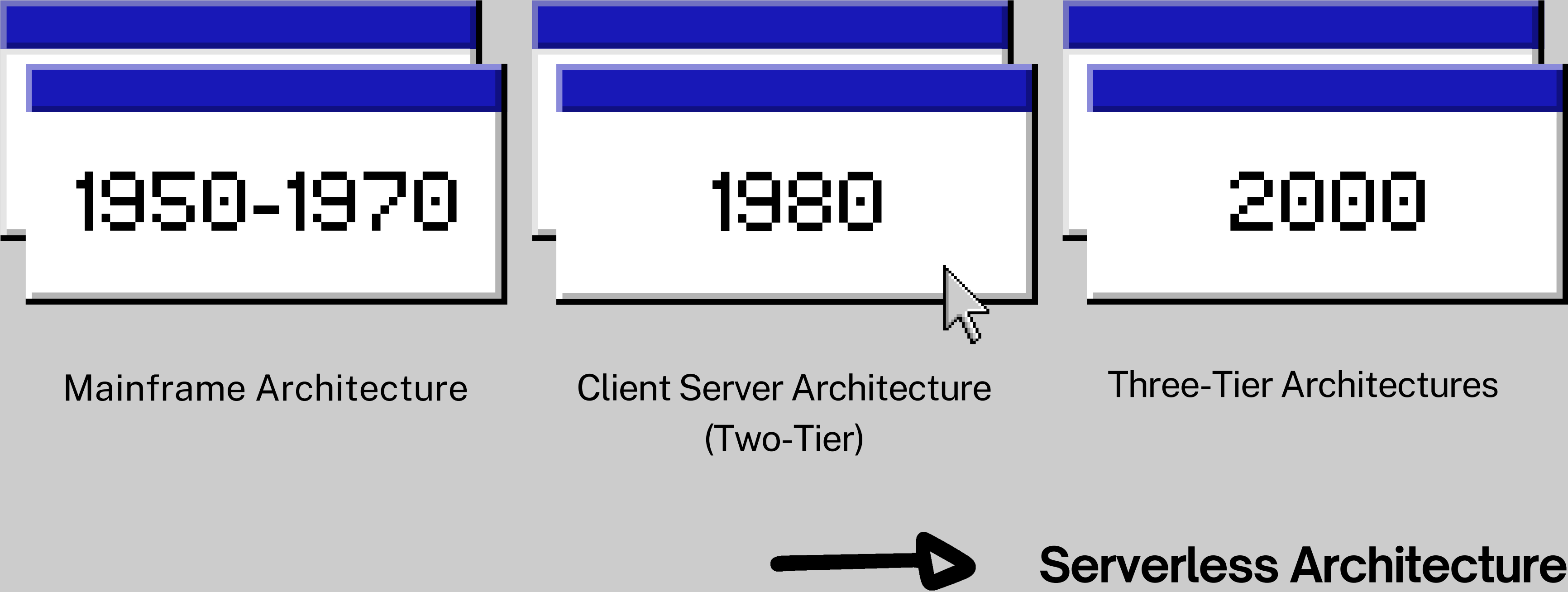
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Timeline

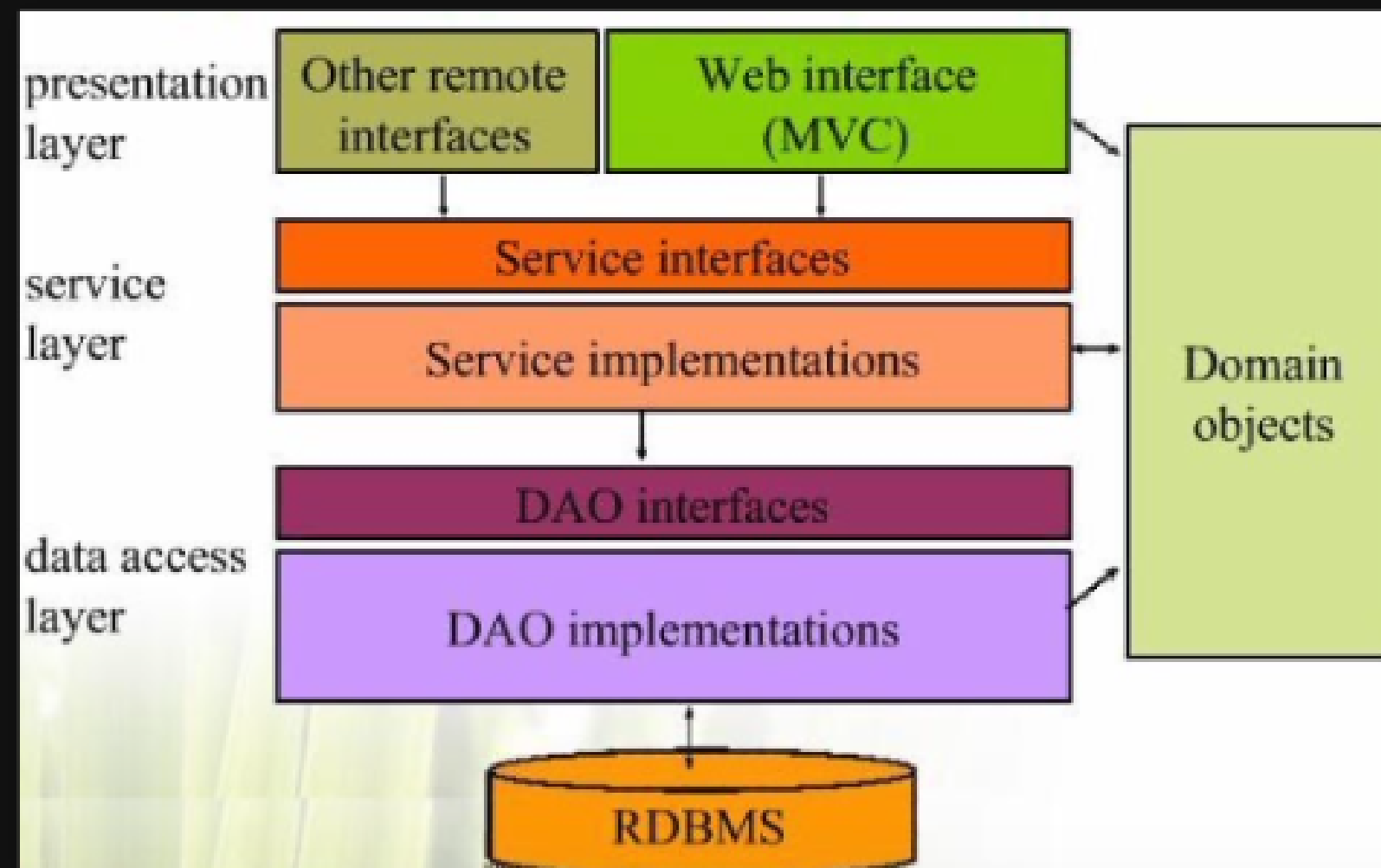


Application Layering



Layering is one of the most common techniques used to break apart a complicated software system

- A single layer can be understood as a coherent whole without knowing much about the other layers
- Layers can be substituted with alternative implementations of the same basic services
- Dependencies between layers can be minimized, reducing the overall coupling in the system

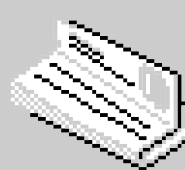
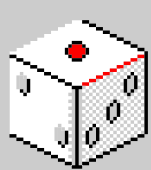
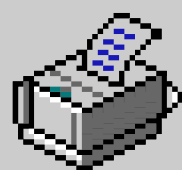
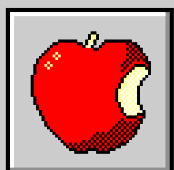
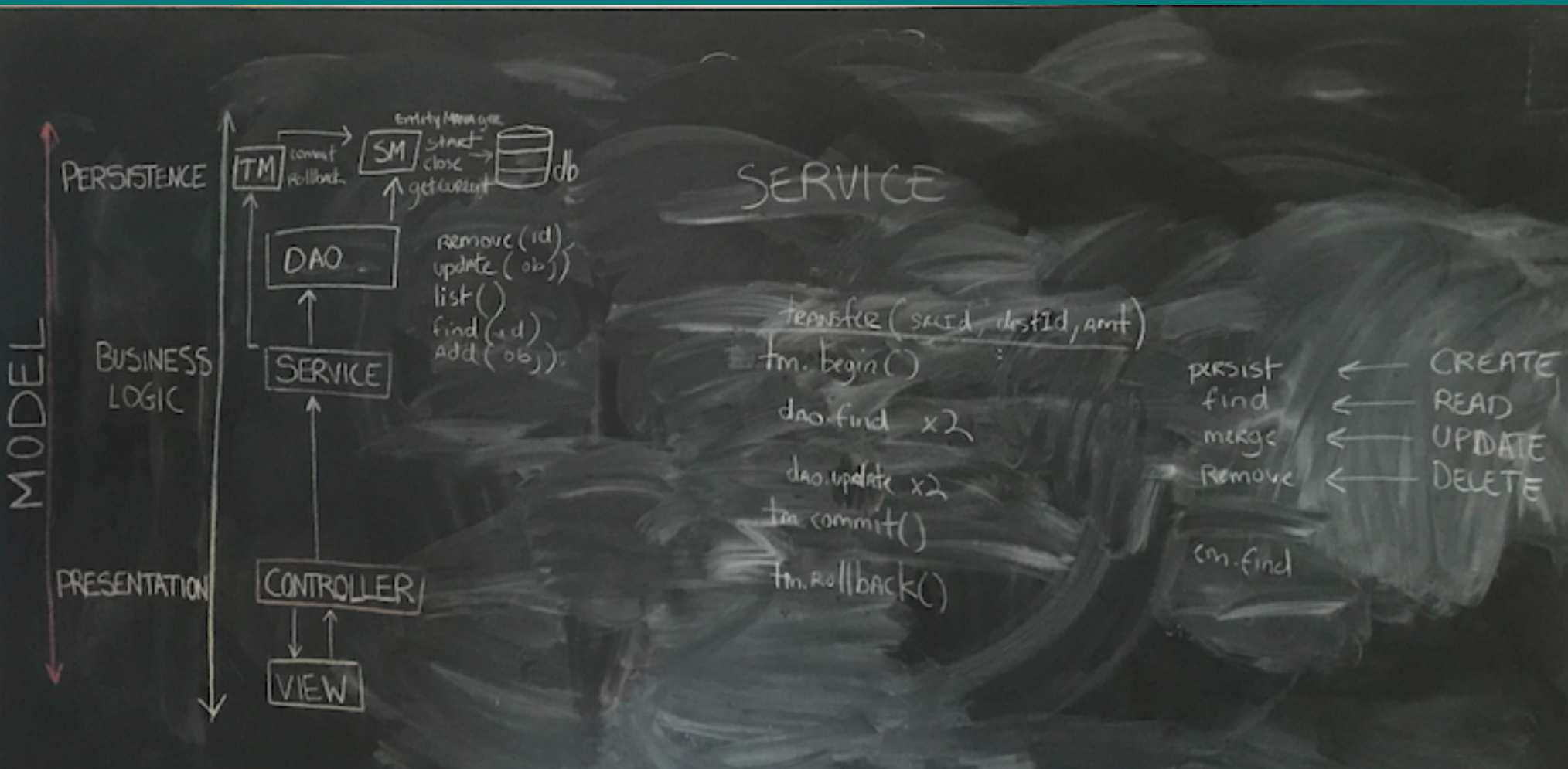


Data Access Object



DAO is an important design pattern used at the persistence layer

- Separates low level data access from high level business services
- Handles data access and manipulation from the persistence storage (db, ldap, file system, etc.)
- Decouples the persistent storage implementation from the rest of the application



Ex.Implement the Generic Dao on the JavaBank application



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```
public class JpaCustomerService implements CustomerService {  
  
    private CustomerDAO dao;  
    private JpaTransactionManager tm;  
  
    public Customer get(Integer id) {  
  
        tm.beginRead();  
        Customer customer = dao.find(id);  
        tm.commit();  
        return customer;  
    }  
  
    public void add(Customer customer){  
        tm.beginWrite();  
        dao.add(customer);  
        tm.commit();  
    }  
}
```





